

Diagonal universality and a Hilbert–Schmidt sufficient class for Problem 1.5 of arXiv:2302.12857

Candidate partial result from run `fa_banach_001`

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Status: candidate partial result; human review required. The packet does not claim a full characterization for arbitrary bounded operators. It proves (i) that the natural diagonal form with arbitrary bounded operator is already universal for all bounded sequences, and (ii) that Hilbert–Schmidt operators give genuine correlation sequences, with a stronger two-variable commuting-action realization.

1 Source and exact question

The source is Or Shalom, *An application of Grothendieck theorem to the theory of multicoloration sequences, multiple recurrence and partition regularity of quadratic equations*, arXiv:2302.12857v1, submitted 24 February 2023 [1]. The question is Problem 1.5 in Section 1.1, printed on page 4 of the source PDF. It refers to the right-hand side of equation (2) from Theorem 1.2 on page 3:

$$\int_X T_\gamma S_{\gamma'} f T_\gamma g h \, d\mu = \int_\Sigma (G\xi_\gamma)(\chi) \xi_{\gamma'}(\chi) \, d\lambda(\chi), \quad \xi_\gamma(\chi) = \chi(\gamma).$$

Theorem 1.2 (Structure theorem for 2-step correlation sequences for non-commutative transformations). *Let $X = (X, \mathcal{B}, \mu)$ be a probability space and let Γ be a countable abelian group. Let $T, S : \Gamma \rightarrow \text{Aut}(X)$ be two (not necessarily commuting) Γ -actions on X . Then for every $f, h \in L^2(X)$ and $g \in L^\infty(X)$, there exists a probability measure λ on the Pontryagin dual $\Sigma := \widehat{\Gamma}$ of Γ , and a continuous operator $G : L^2(\Sigma, \lambda) \rightarrow L^2(\Sigma, \lambda)$ such that*

$$(2) \quad \int_X T_\gamma S_{\gamma'} f \cdot T_\gamma g \cdot h \, d\mu = \int_\Sigma G(\xi_\gamma)(\chi) \cdot \xi_{\gamma'}(\chi) d\lambda(\chi),$$

where $\xi_\gamma : \Sigma \rightarrow S^1$ is the evaluation map $\xi_\gamma(\chi) = \chi(\gamma)$.

Figure 1: Equation (2) and its hypotheses, rendered from page 3 of the source PDF.

Problem 1.5. Determine for which λ and G the expression on the right hand side of (2) is a multicorrelation sequence for commuting T and S , or for $T = S$.

Figure 2: Problem 1.5, rendered from page 4 of the source PDF.

Transcription. “Determine for which λ and G the expression on the right hand side of (2) is a multicorrelation sequence for commuting T and S , or for $T = S$.”

The displayed expression has two independent variables γ, γ' , but the source definition of a multicorrelation sequence has one variable. We therefore separate two precise statements: a result for the full function on Γ^2 , and a result for the natural diagonal $\gamma' = \gamma$. This scope distinction is essential to the claims below.

2 Definitions and conventions

Let Γ be a countable discrete abelian group and let $\Sigma = \widehat{\Gamma}$ be its compact Pontryagin dual. Write $\xi_\gamma(\chi) = \chi(\gamma)$. A k -step multicorrelation sequence in the source is a function

$$c(\gamma) = \int_X \prod_{j=0}^k R_{j\gamma} f_j d\mu, \quad f_j \in L^\infty(X),$$

for a probability-preserving Γ -action R . In particular, an ordinary correlation $\int_X h R_\gamma f d\mu$ is a 1-step multicorrelation and, by adjoining factors equal to 1, is a k -step multicorrelation for every $k \geq 1$.

The source uses the bilinear pairing $\int uv d\lambda$, without complex conjugation. We use exactly this convention. A function on Γ^d is a Fourier–Stieltjes transform if it has the form

$$\gamma \mapsto \int_{\widehat{\Gamma^d}} \omega(\gamma) d\nu(\omega)$$

for a finite complex Borel measure ν .

3 Proof Intuition

There are two complementary mechanisms. For Haar measure, the evaluation characters form an orthonormal basis. A weighted flip of this basis sends ξ_γ to $a(\gamma)\xi_{-\gamma}$, so the bilinear diagonal coefficient is exactly an arbitrary prescribed bounded sequence $a(\gamma)$. Thus mere boundedness of G cannot itself encode the desired dynamical structure.

For a Hilbert–Schmidt operator the situation changes: the operator has an L^2 kernel $K(\chi, \eta)$. Expanding the coefficient turns the entire two-variable expression into the Fourier transform of the finite measure $K(\chi, \eta)d\lambda(\chi)d\lambda(\eta)$ on Σ^2 . A one-circle skew product then realizes every such Fourier transform as an ordinary correlation of two commuting actions. On the diagonal the two actions combine into one, giving a multicorrelation sequence in the source’s precise sense. The remaining open territory is the bounded non-Hilbert–Schmidt case and any desired intrinsic characterization of those pairs (λ, G) .

4 Candidate partial theorem

Theorem 1 (Universality versus the Hilbert–Schmidt class). *Let Γ be a countable discrete abelian group, $\Sigma = \widehat{\Gamma}$, and $\xi_\gamma(\chi) = \chi(\gamma)$.*

- (a) **Diagonal universality.** *Let m be normalized Haar measure on Σ . For every $a \in \ell^\infty(\Gamma)$ there is a bounded linear operator $G_a : L^2(\Sigma, m) \rightarrow L^2(\Sigma, m)$ with*

$$\|G_a\| = \|a\|_\infty, \quad \int_{\Sigma} (G_a \xi_\gamma)(\chi) \xi_\gamma(\chi) dm(\chi) = a(\gamma)$$

for all $\gamma \in \Gamma$. Conversely, every diagonal coefficient produced by a probability measure λ and a bounded G is bounded, with supremum norm at most $\|G\|$. Hence, when arbitrary pairs (λ, G) are allowed, the diagonal forms are exactly $\ell^\infty(\Gamma)$.

- (b) **A two-variable sufficient class.** *Let λ be any Borel probability measure on Σ and let $G : L^2(\Sigma, \lambda) \rightarrow L^2(\Sigma, \lambda)$ be Hilbert–Schmidt. Then*

$$b_G(\gamma, \gamma') := \int_{\Sigma} (G \xi_\gamma)(\chi) \xi_{\gamma'}(\chi) d\lambda(\chi)$$

is a Fourier–Stieltjes transform on Γ^2 . More strongly, there are a probability space (Y, μ) , two commuting probability-preserving Γ -actions T, S , and $f, h \in L^\infty(Y)$ such that

$$b_G(\gamma, \gamma') = \int_Y T_\gamma S_{\gamma'} f T_\gamma 1 h d\mu = \int_Y T_\gamma S_{\gamma'} f h d\mu.$$

- (c) **Diagonal consequence.** *The diagonal $a_G(\gamma) = b_G(\gamma, \gamma)$ is an ordinary correlation sequence for a single probability-preserving Γ -action. It is therefore a multicorrelation sequence under the source definition. If K is the L^2 kernel of G , the representing Fourier–Stieltjes measure can be chosen with total variation at most*

$$\|K\|_{L^1(\lambda \otimes \lambda)} \leq \|K\|_{L^2(\lambda \otimes \lambda)} = \|G\|_{\text{HS}}.$$

Proof. Part (a). By Pontryagin duality and Haar orthogonality, the characters $\{\xi_\gamma : \gamma \in \Gamma\}$ form an orthonormal basis of $L^2(\Sigma, m)$ [2, Chapter 1]. Define on this basis

$$G_a \xi_\gamma = a(\gamma) \xi_{-\gamma}.$$

The map $\gamma \mapsto -\gamma$ is a permutation of Γ , so for every finitely supported scalar family (c_γ) ,

$$\left\| G_a \sum_{\gamma} c_\gamma \xi_\gamma \right\|_2^2 = \sum_{\gamma} |a(\gamma) c_\gamma|^2 \leq \|a\|_\infty^2 \sum_{\gamma} |c_\gamma|^2.$$

Thus G_a extends to a bounded linear operator. Testing it on basis vectors whose weights approach the supremum gives $\|G_a\| = \|a\|_\infty$. Finally,

$$\int_{\Sigma} (G_a \xi_\gamma) \xi_\gamma dm = a(\gamma) \int_{\Sigma} \xi_{-\gamma} \xi_\gamma dm = a(\gamma).$$

In the other direction, because $|\xi_\gamma| = 1$ and λ is a probability measure, Cauchy–Schwarz gives

$$\left| \int_{\Sigma} (G \xi_\gamma) \xi_\gamma d\lambda \right| \leq \|G \xi_\gamma\|_2 \|\xi_\gamma\|_2 \leq \|G\|.$$

This proves the exact universality assertion.

Part (b): kernel and measure. The Hilbert–Schmidt kernel identification gives $K \in L^2(\Sigma \times \Sigma, \lambda \otimes \lambda)$ such that

$$(Gu)(\chi) = \int_{\Sigma} K(\chi, \eta) u(\eta) d\lambda(\eta)$$

for $u \in L^2(\Sigma, \lambda)$, with $\|K\|_2 = \|G\|_{\text{HS}}$ [3, Chapter 2]. Since λ is a probability measure, $K \in L^1$. Fubini’s theorem yields

$$\begin{aligned} b_G(\gamma, \gamma') &= \iint_{\Sigma^2} K(\chi, \eta) \eta(\gamma) \chi(\gamma') d\lambda(\eta) d\lambda(\chi) \\ &= \int_{\Sigma^2} \eta(\gamma) \chi(\gamma') d\nu(\eta, \chi), \end{aligned}$$

where

$$d\nu(\eta, \chi) = K(\chi, \eta) d\lambda(\eta) d\lambda(\chi).$$

This is a Fourier–Stieltjes transform on Γ^2 , and $\|\nu\| \leq \|K\|_1 \leq \|K\|_2$.

Part (b): commuting dynamical realization. Write $d\nu = q d\rho$, where $\rho = |\nu|$, $|q| = 1$ ρ -almost everywhere, and let $M = \rho(\Sigma^2)$. The case $M = 0$ is immediate, so assume $M > 0$. On

$$Y = \Sigma \times \Sigma \times \mathbb{T}, \quad \mu = \frac{\rho}{M} \otimes m_{\mathbb{T}},$$

define point transformations

$$\tau_{\gamma}(\eta, \chi, z) = (\eta, \chi, \eta(\gamma)z), \quad \sigma_{\gamma'}(\eta, \chi, z) = (\eta, \chi, \chi(\gamma')z).$$

They are commuting probability-preserving Γ -actions: they fix the base and rotate the Haar circle in the fibre. Let their Koopman operators be $T_{\gamma}F = F \circ \tau_{\gamma}$ and $S_{\gamma'}F = F \circ \sigma_{\gamma'}$. Put

$$f(\eta, \chi, z) = z, \quad h(\eta, \chi, z) = Mq(\eta, \chi)\bar{z}.$$

Both functions are bounded. Direct substitution gives

$$\begin{aligned} \int_Y T_{\gamma} S_{\gamma'} f h d\mu &= \int_{\Sigma^2 \times \mathbb{T}} \eta(\gamma) \chi(\gamma') z Mq(\eta, \chi) \bar{z} d(\rho/M) dm_{\mathbb{T}} \\ &= \int_{\Sigma^2} \eta(\gamma) \chi(\gamma') d\nu(\eta, \chi) = b_G(\gamma, \gamma'). \end{aligned}$$

Taking the middle function in the source’s equation (2) to be 1 supplies the claimed full two-variable realization.

Part (c). Because T and S commute, $R_{\gamma} = T_{\gamma} S_{\gamma}$ is a probability-preserving Γ -action. Consequently

$$a_G(\gamma) = b_G(\gamma, \gamma) = \int_Y R_{\gamma} f h d\mu$$

is an ordinary correlation. It matches the source definition with $f_0 = h$, $f_1 = f$ and, for any desired larger step, all remaining functions equal to 1. Equivalently, its Fourier–Stieltjes measure is the pushforward of ν by $(\eta, \chi) \mapsto \eta\chi$. The norm estimate was established above. $\square$

5 Consequences and precise limitations

1. Every finite-rank, trace-class, or Hilbert–Schmidt G is covered by Theorem 1(b). In particular, if $L^2(\Sigma, \lambda)$ is finite dimensional, every bounded G is covered.
2. Part (a) is a sharp obstruction to a characterization based only on operator norm or generic boundedness: with the admissible Haar measure, the diagonal construction produces every bounded sequence.
3. The theorem does *not* characterize which arbitrary bounded, non-Hilbert–Schmidt operators give multicorrelation sequences. Nor does it resolve whether a particular bounded sequence is a multicorrelation sequence; the source itself identifies that classification as a major open problem.
4. The phrase “for commuting T and S ” in Problem 1.5 may intend a specific one-variable specialization other than the diagonal. The packet’s full Γ^2 realization is unambiguous and stronger at the level of the displayed two-variable coefficient; its formal multicorrelation conclusion is stated only for the diagonal.

6 Verification notes

The accompanying `verification.md` contains an explicit adversarial verifier report. The main checks are:

1. the weighted Fourier-basis flip is complex-linear and remains valid for groups with 2-torsion;
2. the source’s bilinear (not sesquilinear) pairing is used throughout;
3. the order of the kernel variables gives exactly $\eta(\gamma)\chi(\gamma')$;
4. the skew-product transformations satisfy the group laws, commute, and preserve the probability measure;
5. the polar-density factor M is placed in h , so the correlation has the correct normalization and both observables are bounded;
6. setting the source’s middle observable to 1 reproduces equation (2), and combining the commuting actions proves the diagonal claim.

No numerical experiment is used as proof. The only code in the packet renders the two source crops from the copied source PDF.

7 Bounded novelty check and review recommendation

On 19 July 2026, the run’s solution, attempt, proof-gap, and ledger indexes were searched for arXiv:2302.12857 and for the core phrases “Grothendieck multicorrelation”, “for which λ and G ”, “Hilbert–Schmidt”, and “Fourier–Stieltjes”; no duplicate result packet was found. Exact-title and exact-problem web searches, the arXiv record, an author follow-up search, and a bounded OpenAlex citation query did not locate a later paper explicitly answering Problem 1.5. Semantic Scholar rate-limited the query. A recent author paper, arXiv:2607.13286, concerns a different non-vanishing question. These checks are bounded and do not establish originality.

Human-review recommendation. Treat this as a likely valid, substantial partial result. The highest-value review points are (i) the intended one-variable reading of Problem 1.5, (ii) the Hilbert–Schmidt kernel convention for the source’s bilinear pairing, and (iii) whether an equivalent sufficient class or the diagonal-universality observation already appears under different terminology. Do not promote it to a full solution without a characterization of the arbitrary bounded case.

References

- [1] O. Shalom, *An application of Grothendieck theorem to the theory of multicorrelation sequences, multiple recurrence and partition regularity of quadratic equations*, arXiv:2302.12857v1 (2023), <https://arxiv.org/abs/2302.12857>.
- [2] W. Rudin, *Fourier Analysis on Groups*, Interscience Publishers, 1962.
- [3] B. Simon, *Trace Ideals and Their Applications*, 2nd ed., American Mathematical Society, 2005.